Timothy Lasater

timothy.a.lasater@gmail.com • (949) 291-3592 • Irvine, CA • timothylasater.com

EDUCATION

University of California, Irvine **June 2026**
Bachelor of Science, **Biomedical Engineering** | Summa Cum Laude | Phi Beta Kappa | GPA **3.986**
Specialization in Micro/Nano-scale Engineering

SKILLS

Software: SolidWorks (**CSWA** Certified), Ansys (CFD, FEA), MATLAB, Python, JMP, C (Arduino), Minitab
Process & Quality: Design of Experiments (DOE), Lean Six Sigma, DMAIC, 5S, Project Management
Lab & Fabrication: 3D Printing (FDM, SLA), Photolithography, Soft Lithography, Breadboarding, DAQ, PCR

EXPERIENCE

Research Associate / Device Engineer, Scientific Horizons Consulting **October 2024 - Present**

- Lead design and prototyping of portable nebulizer devices with novel actuation and drug delivery mechanisms in SolidWorks, leading development of **>25** device iterations to optimize delivery
- Lead Design of Experiments (**DOE**) for device evaluation; analyze results in **Python, JMP, & Excel**
- Construct benchtop circuits & test fixtures to validate nebulizer designs against commercial competitors, measuring airflow pressure amplitude & targeted aerosol delivery efficiency
- Write Standard Operating Procedures (**SOP**) for novel & improved experimental protocols
- Conduct benchtop testing to evaluate oral drug dissolution and trans-buccal permeation using custom-designed, 3D printed simulated oral environment, informing formulation improvements

Team Lead, PROMPTly: Integrated Test for Amniotic Fluid Detection **February 2025 - June 2026**

- Led design of an at-home diagnostic pad integrating a lateral flow immunoassay (LFT) for detecting biomarkers of Premature Rupture of Membranes (PROM) in amniotic fluid
- Developed and tested fluid-routing and volume-control mechanisms to achieve controlled flow
- Optimized material and flow paths with synthetic fluid analog to validate sensitivity & specificity
- Directed development, prototyping & market analysis, considering user needs and manufacturing
- Defined regulatory considerations, IP pathway, business plan, and go-to-market strategy

Undergraduate Researcher, Hui Lab, University of California, Irvine **June 2024 - June 2026**

- Designed and constructed an automated droplet pump for combinatorial drug screening with computer vision tracking, reducing experimental duration and error rate by up to 70%
- Secured **$1,500** in funding and presented at the UCI Undergraduate Research Symposium
- Simulated and optimized microfluidic flow and mixing using CFD software, improving model accuracy by more than **50%** and real-life mixing performance by over **33%**
- Optimized low-cost fabrication process to expedite microfluidic design-build-test cycle by **90%**, which was implemented in a microfluidics lab class at UCI with >60 students

Manufacturing Engineering Intern, Medtronic Cardiac Surgery **June 2025 - August 2025**

- Applied Lean Six Sigma methodologies, including DMAIC, Gemba observation and statistics (Minitab), to engineer a solution to a longstanding, recurring product quality defect (25+ years)
- Spearheaded implementation of a scalable process improvement potentially reducing defect rate by **20%**, cycle time by **75%**, and with projected annual savings of **$50,000**
- Collaborated with R&D, Quality, and line operators to coordinate testing protocols and accelerate verification and validation timelines by two weeks, ensuring seamless handoff after internship